

Internship in Physical Organic Chemistry

Week 2 — Part 4: Writing the Experimental Chapter

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Goals of this part:

- Understand the structure and function of an Experimental chapter
- Write claim sentences that define the outline before drafting
- Draft each paragraph from your scratch notes, not from memory
- Revise for structure and then for sentence quality
- Complete scratch sections 2W-exp and 2W-f

The Experimental chapter is the most constrained section of a scientific report. Its function is precise: to describe what was done in sufficient detail that another scientist could reproduce the experiment. It is written in the past tense, uses passive or active voice consistently, and contains no interpretation, no results, and no discussion. Every sentence must be traceable to something you actually did.

1 What belongs in an Experimental chapter

An Experimental chapter for an MS/MS study contains the following elements, always in this order:

1. **Instrument description.** Name the instrument (manufacturer, model). Describe the ion source and the ion path briefly: how ions are formed, how they are transferred, how they are fragmented, and how they are detected. State the fragmentation method (HCD) and the mass analyser (Orbitrap). State the software used for data acquisition, including the version number.
2. **Sample preparation.** Name the compound (systematic name or common name with molecular formula). State the solvent, the concentration, and any handling precautions relevant to the stability of the compound.
3. **Source conditions.** State the spray voltage (positive and negative mode separately). State the in-source voltage parameter and the value used. State any other source parameters that were set non-standardly.
4. **Acquisition parameters.** State the scan range (m/z range), the resolution setting, the isolation window for MS/MS, and the NCE values used. State the number of scans averaged if relevant.
5. **Statement of what was recorded.** A brief sentence stating which spectra were acquired: full-scan MS1 in positive and negative mode, and MS/MS spectra for which precursor ions.

The Experimental chapter does not explain *why* you chose these conditions. Justification belongs in the Results and Discussion. Here you state only what was done. The one exception is sample handling precautions (e.g. light sensitivity): these are reported in the Experimental chapter because they are part of the procedure, not because they require justification.

2 Phase A — Outline

Before writing any prose, write one claim sentence for each element of the Experimental chapter listed above. A claim sentence in a Methods section is a factual statement of what was done, written as a complete sentence.

The purpose of writing claim sentences first is to force you to decide, before touching the prose, exactly what information each paragraph will contain. This prevents two common errors: writing vague paragraphs that omit key parameters, and writing bloated paragraphs that mix instrument description with sample preparation.

Write your claim sentences in scratch section 2W-exp now, before continuing. Use the factual inventory you recorded there during acquisition as your source. Do not write from memory — write from your notes. Every parameter that will appear in the Experimental chapter must already be recorded in 2W-exp. If something is missing, check your lab journal.

Example claim sentences for a generic chemistry experiment (SN2 reaction):

Instrument: The reaction was monitored by ^1H NMR spectroscopy on a 400 MHz spectrometer (Bruker Avance III, Bruker BioSpin, Rheinstetten, Germany).

Sample: A solution of 1-bromobutane (0.10 mmol) in deuterated acetone (0.6 mL) was prepared immediately before each measurement.

Conditions: Spectra were recorded at 298 K with 16 scans per acquisition.

Parameters: The spectral window was set to 10 ppm, with a relaxation delay of 2 s between scans.

What was recorded: Full ^1H NMR spectra were recorded at 5 min intervals over 2 h to monitor the disappearance of the starting material and the appearance of the product.

Note how each sentence contains exactly one piece of information, is written in the past tense, and contains no interpretation. The instrument sentence gives manufacturer and model. The sample sentence gives identity, amount, and solvent. The conditions sentence gives temperature and number of scans.

3 Phase B — Drafting

Once your claim sentences are written and checked against your lab journal, expand each one into a short paragraph. For an Experimental chapter, paragraphs are typically 2–4 sentences. Each paragraph covers one element from the list in Section 1.

3.1 Drafting the instrument description paragraph

The instrument description paragraph has a standard structure in MS papers: instrument name and manufacturer, followed by a brief description of the ion path. Write it as a narrative that follows the ions from source to detector.

Structure of an instrument description paragraph for an MS/MS experiment:

Opening sentence: Name the instrument with full manufacturer information.

Middle sentences: Describe the ion path in order — ionisation, desolvation, ion transfer, precursor selection, fragmentation, detection.

Closing sentence: State the acquisition software and version number.

Template (replace all bracketed items with your actual values):

All measurements were performed on a [instrument model] hybrid mass spectrometer ([manufacturer], [city], [country]). Ions were generated by electrospray ionisation (ESI) and transferred into the instrument through heated ion transfer optics; desolvation was assisted by a nitrogen sheath gas flow. Precursor ions were isolated using the quadrupole mass filter and fragmented by higher-energy collisional dissociation (HCD) in the collision cell; product ions were detected in the Orbitrap mass analyser at a resolving power of [X] FWHM at m/z [Y]. Data were acquired and processed using [software name], version [X.X] ([manufacturer], [city], [country]).

Write your own version using the actual instrument name, parameters, and software version from your lab journal and scratch section 2W-exp. Do not copy this template word for word — use it as a structural guide only. The wording of your paragraph must reflect your own instrument and your own experiment.

3.2 Drafting the sample preparation paragraph

The sample preparation paragraph must include: compound name and molecular formula, solvent, concentration, and any stability-related handling precautions. It should be 2–3 sentences.

The solvent choice and concentration are reported as facts here. The justification for these choices does not belong in the Experimental chapter. If you want to note that the choice was guided by literature precedent, that sentence belongs in the Results and Discussion, not here.

3.3 Drafting the source conditions paragraph

Report the spray voltage for positive and negative mode separately. Report the in-source voltage parameter by its exact name as it appears in Xcalibur, and the value used. Report only the final values used for acquisition — the optimisation process is not described in the Experimental chapter.

3.4 Drafting the acquisition parameters paragraph

Report: scan range, resolution, isolation window, and NCE values. If you used different NCE values for different precursors, state this explicitly. The number of spectra recorded per precursor and the number of scans averaged should also be stated if non-standard.

3.5 Drafting the statement of what was recorded

This is typically one or two sentences. State which compounds were measured, in which ionisation modes, and which precursor ions were selected for MS/MS. Keep it factual and complete.

4 Phase C — Revision

Revision of an Experimental chapter follows the same two-pass approach as Week 1: a structure pass first, then a sentence pass.

4.1 Structure pass

Extract the first sentence of each paragraph and read them in sequence. Ask:

- Does each paragraph cover exactly one element from the list in Section 1?

- Is the order correct (instrument → sample → source → parameters → what was recorded)?
- Is any element missing?
- Is any element repeated across two paragraphs?

4.2 Sentence pass

Read each sentence and apply the following checks:

- **Past tense throughout.** Experimental chapters are written in the past tense: *spectra were recorded*, not *spectra are recorded*.
- **No interpretation.** Remove any sentence that explains why a parameter was chosen, what the result was, or what the observation means. These belong in Results and Discussion.
- **Every parameter is stated with its value and unit.** *The spray voltage was set to 3.5 kV*, not *the spray voltage was optimised*.
- **Manufacturer information is complete.** The first mention of any instrument or software must include manufacturer name, city, and country in parentheses.
- **Abbreviations are defined on first use.** ESI, HCD, NCE, and FWHM must be spelled out the first time they appear, with the abbreviation in parentheses immediately after.

Common errors in Experimental chapters — examples from the SN2 domain:

Interpretation in Methods (incorrect):

The reaction was run in acetone because acetone is a polar aprotic solvent that does not compete with the nucleophile.

→ Remove the justification. Write: *The reaction was performed in acetone.*

Missing units (incorrect):

The temperature was set to room temperature.

→ Write: *The reaction was performed at 298 K.*

Vague parameter (incorrect):

A standard concentration was used.

→ Write: *Solutions were prepared at a concentration of $1.0 \times 10^{-4} \text{ mol L}^{-1}$.*

Incomplete manufacturer information (incorrect):

Spectra were recorded on a Bruker spectrometer.

→ Write: *Spectra were recorded on an Avance III spectrometer (Bruker BioSpin, Rheinstetten, Germany).*

5 Completing the working log

Once the Experimental chapter draft is complete, fill in scratch section 2W-f: write one sentence stating whether your experimental observations are consistent with what your Background chapter predicted, and one sentence identifying the most important open question that Week 3 will address.

This is the bridge between Week 2 and Week 3. Writing it forces you to connect the data you have just collected to the physical organic chemistry question at the centre of the project.

Submit your completed Experimental chapter draft in Overleaf before the end of Week 2. Your supervisor will read both the chapter and your `scratch_week02.tex` log. The draft does not need to be perfect — it needs to be complete. A complete draft with some imprecision is more useful than a polished fragment.